

# **Development of an Energy Management Strategy for a Plug-in Fuel Cell Hybrid Electric Vehicle (FCHEV)**

**MAT 4901 — Graduation Project I**

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## 1. Definition and Purpose of Design

The purpose of this study is to establish, from first principles, the complete mathematical formulation of an optimal Energy Management Strategy (EMS) that minimises the operational cost of a plug-in hydrogen fuel cell hybrid electric vehicle (FCHEV).

Hydrogen fuel cell vehicles are regarded as one of the principal pathways towards the decarbonisation of road transport, since the only tailpipe emission of a proton exchange membrane fuel cell (PEMFC) is water vapour. A fuel cell stack alone, however, is poorly suited to direct traction duty: its dynamic response to rapid load transients is slow, it cannot absorb the energy recovered during regenerative braking, and frequent load cycling accelerates membrane degradation [1]. For these reasons, the fuel cell is hybridised with a lithium-ion battery pack, and the resulting architecture contains two independent energy sources. In the plug-in configuration considered in this study, the battery is additionally charged from the electrical grid prior to the trip, so that the stored electrochemical energy originates from a source whose monetary cost differs fundamentally from that of the hydrogen carried on board.

Whenever the driver demands a traction power  $P_{load}(t)$ , a supervisory controller must decide, at every instant, what fraction of this demand is supplied by the fuel cell system and what fraction is supplied by the battery. This decision rule is the Energy Management Strategy, and its quality directly determines the hydrogen consumption, the electricity consumption, the operating cost and the lifetime of the powertrain components [1], [2]. The problem is naturally cast as a constrained optimal control problem in which a cumulative cost functional is minimised subject to the power balance of the DC bus, the state-of-charge (SoC) dynamics of the battery and the physical limits of the components.

The work is distributed over two graduation project terms. Within the scope of Graduation Project I, which is reported here, the system models of the reference vehicle, the PEMFC stack and the battery pack are formulated mathematically, the operational cost function is derived for the plug-in charge-depleting scenario, the constraints are defined, and the optimal control problem is stated in its complete mathematical form together with the formulation of four solution strategies. Within the scope of Graduation Project II, this formulation will be solved numerically in a Python-based simulation environment using Dynamic Programming (DP), Adaptive Equivalent Consumption Minimization Strategy (A-ECMS), Model Predictive Control (MPC) and Boundary Layer Surface Following (BLFS), and the strategies will be compared quantitatively.

## 2. Scope of Design and Areas of Usage

The study is conducted in a sequence of well-defined steps. First, a literature review is carried out on energy management strategies for fuel cell hybrid vehicles, on zero-

dimensional PEMFC modelling and on lumped-parameter equivalent circuit models of lithium-ion batteries. Second, the system architecture is fixed: the T4 topology, in which the PEMFC is connected to the DC bus through a unidirectional DC/DC converter while the battery is connected directly, is adopted, the second-generation Toyota Mirai is selected as the reference vehicle, and two consecutive Worldwide Harmonized Light Vehicles Test Procedure (WLTP) Class 3b cycles are selected as the reference trip. Third, the mathematical models of the powertrain components are constructed: the longitudinal vehicle dynamics, the Mann–Amphlett Generalised Steady-State Electrochemical Model (GSSEM) of the fuel cell stack, and the first-order resistor–capacitor equivalent circuit model with hysteresis (1-RC + Hysteresis) of the battery. Fourth, the system-wide power flow model is assembled and the conversion chains from the control decision to the hydrogen mass flow rate and to the battery state of charge are derived. Fifth, the operational cost function combining the hydrogen cost and the grid electricity cost is derived for the charge-depleting plug-in scenario, parametrised with Germany 2026 price scenarios. Finally, the complete optimal control problem is stated and the four EMS strategies to be implemented in the second term are formulated.

The outputs of the study can be used in several areas. The formulation provides a directly transferable basis for the supervisory control software of plug-in fuel cell passenger cars, and, with modified parameter sets, of fuel cell city buses, light commercial vehicles and material handling equipment. The cost function, being parametrised with current European hydrogen and electricity prices, can further be employed in techno-economic feasibility analyses of fuel cell mobility in the European market, and the reproducible parameter catalogue assembled from primary sources can serve as a benchmark configuration for subsequent academic studies.

## **3. Conducted Studies**

### **3.1. Literature Review**

#### **3.1.1. Energy management strategies for fuel cell hybrid vehicles**

Energy management strategies reported in the literature are commonly grouped into three classes [1]. Rule-based strategies, comprising deterministic rules, state machines and fuzzy logic controllers, encode expert knowledge in the form of power-split heuristics; they are computationally trivial and robust, but they carry no optimality guarantee. Optimization-based strategies formulate the power split as an optimal control problem. Dynamic Programming computes the global optimum by backward recursion over the whole drive cycle and is therefore usable only offline, as a benchmark [15], [16]. Pontryagin’s Minimum Principle (PMP) converts the constrained problem into the pointwise minimisation of a Hamiltonian, and the Equivalent Consumption Minimization Strategy (ECMS) can be interpreted as a real-time realisation of PMP in which the co-state is replaced by an

equivalence factor [2], [3]. Xu et al. applied PMP to a plug-in fuel cell city bus and demonstrated that the PMP solution remains within approximately 1.4 % of the DP optimum while being implementable in real time; the same study proved that a classical ECMS with a constant equivalence factor may degenerate into a charge-depleting/charge-sustaining (CDCS) strategy in the plug-in case [4]. Adaptive variants of ECMS therefore update the equivalence factor online, typically by a proportional-integral feedback on the SoC tracking error [5]. Model Predictive Control solves a finite-horizon optimisation at every sampling instant and applies only the first element of the resulting control sequence [1]. Learning-based strategies, including deep reinforcement learning, have recently been surveyed in [1]; they are excluded from the present scope because their training requirements are incompatible with the available data.

### 3.1.2. PEMFC modelling approaches

Fuel cell models are classified by spatial resolution into zero- to three-dimensional models, and by modelling philosophy into mechanistic, semi-empirical and data-driven models [6], [7]. For EMS-oriented studies, zero-dimensional semi-empirical models are the established choice, since the controller requires only the static input-output map between stack power, efficiency and hydrogen consumption. The Generalised Steady-State Electrochemical Model of Mann and Amphlett [6] expresses the cell voltage as the Nernst potential reduced by activation, ohmic and concentration losses, building upon the earlier polarisation models of Springer et al. [8] and Kim et al. [9]. Compared with dynamic zero-dimensional models such as that of Ziogou et al. [10], *a static formulation requires only algebraic evaluations; an operation-count analysis carried out within the present study indicates that this reduces the computational load of a DP optimisation by more than two orders of magnitude at the expense of a steady-state error of only a few per cent, which is the decisive argument for its adoption here.*

### 3.1.3. Lithium-ion battery modelling approaches

Battery models suitable for real-time state estimation fall into electrochemical, data-driven and equivalent-circuit categories. Nejad et al. compared ten lumped-parameter equivalent circuit models systematically on  $\text{LiFePO}_4$  (LFP) and  $\text{LiNMC}$  cells and reported that the first-order RC model augmented with a hysteresis state (1-RC + Hysteresis) achieves the lowest SoC estimation error for LFP chemistry (1.39 % mean error), together with marked robustness against erroneous SoC initialisation [11]. The hysteresis description follows Huria et al. [12], and the underlying state and parameter estimation framework is that of Plett [13]. Because the LFP open-circuit-voltage curve is almost flat between 20 % and 80 % SoC, the hysteresis state is indispensable for this chemistry [11], [12]. The two-level modelling practice, in which a high-fidelity plant model is combined with a simplified controller model inside the optimisation layer, follows the established methodology of Onori et al. [14].

### 3.2. System Architecture, Reference Vehicle and Drive Cycle

The architectural selections of the study are summarised in Table 1, where each choice is justified in the section indicated.

*Table 1. Architectural selections of the study.*

| Architectural component | Selection                                                   | Section |
|-------------------------|-------------------------------------------------------------|---------|
| Topology                | T4: PEMFC + unidirectional DC/DC + DC bus + battery         | 3.2     |
| Reference vehicle       | Toyota Mirai II (passenger car)                             | 3.2     |
| Drive cycle             | 2 × WLTP Class 3b (1 h, 46.5 km)                            | 3.2     |
| PEMFC model             | Mann–Amphlett 0-D semi-empirical (GSSEM)                    | 3.3.2   |
| Battery cell / pack     | A123 ANR26650M1B (LiFePO <sub>4</sub> ), 96s10p, ≈ 7.92 kWh | 3.3.4   |
| Battery model           | 1-RC + Hysteresis [11]                                      | 3.3.4   |
| Cost scenario           | Germany 2026: € 11/kg H <sub>2</sub> , € 0.35/kWh           | 3.5     |
| EMS strategies          | DP, A-ECMS, MPC, BLFS                                       | 3.6     |

In the T4 topology, which is the most widely adopted structure in the FCHEV literature, the fuel cell stack feeds the DC bus through a unidirectional DC/DC converter, whereas the battery is connected to the bus directly and exchanges power bidirectionally. The natural decision variable of the EMS is consequently the power delivered by the DC/DC converter to the bus,

$$u(t) \equiv P_{dc}(t), \quad (1)$$

and, once Eq. (1) is fixed, the battery power follows directly from the power balance of the DC bus,

$$P_{bat}(t) = P_{load}(t) - P_{dc}(t), \quad (2)$$

so that a single decision variable fully determines the power split. The sign convention adopted in Eq. (2) is that  $P_{bat} > 0$  denotes battery discharge and  $P_{bat} < 0$  denotes battery charging, either by regenerative braking or by surplus fuel cell power; the unidirectional converter enforces  $P_{dc} \geq 0$ . The only memory of the system is the battery state of charge, which is therefore taken as the state variable,  $x(t) \equiv SoC(t)$ . The hydrogen tank level is

deliberately not treated as a state, because the hydrogen cost enters the running cost directly and no terminal constraint is imposed on the tank; the same simplification is adopted in [4], [5].

The second-generation Toyota Mirai (2020+) is selected as the reference vehicle, since it is the only series-production fuel cell passenger car for which a complete and recent experimental energy balance is available in the literature [17]. The commercial Mirai II is not a plug-in vehicle; in this study its battery is rescaled to approximately 7.92 kWh to provide the plug-in functionality, while all remaining vehicle parameters are kept at their published values. The vehicle parameters used in the dynamic model are listed in Table 2.

*Table 2. Toyota Mirai II reference vehicle parameters [17] and Toyota official technical specifications [21].*

| Parameter                      | Symbol         | Value                      |
|--------------------------------|----------------|----------------------------|
| Total mass (incl. driver)      | $m_v$          | 1925 kg                    |
| Frontal area                   | $A_f$          | 2.547 m <sup>2</sup>       |
| Aerodynamic drag coefficient   | $C_d$          | 0.29                       |
| Rolling resistance coefficient | $C_r$          | 0.012                      |
| Air density (20 °C)            | $\rho_{air}$   | 1.18 kg/m <sup>3</sup>     |
| Drivetrain efficiency          | $\eta_{dt}$    | 0.92                       |
| Maximum motor power            | $P_{m,max}$    | 134 kW                     |
| Stack net peak power           | $P_{fc}^{max}$ | 128 kW (system net 110 kW) |
| DC/DC converter efficiency     | $\eta_{dc}$    | 0.95                       |

The reference trip consists of two consecutive WLTP Class 3b cycles [22], giving a total duration of  $T=3600$  s and a distance of 46.5 km, representative of a typical daily commute in Germany. Two cycles are required so that the 7.92 kWh pack can be depleted by a meaningful amount within the trip. Since the speed profile of the cycle is known in advance, the load trajectory  $P_{load}(t)$  is computable a priori, which is the property exploited by the DP benchmark. The problem is discretised with a sampling period  $\Delta t=1$  s, yielding  $N=3600$  decision instants.

### 3.3. Mathematical Modelling of the Powertrain Components

#### 3.3.1. Longitudinal vehicle dynamics

The power that must be transmitted to the wheels at vehicle speed  $v(t)$  and acceleration  $a(t)=dv/dt$  on a road of grade angle  $\alpha$  is obtained from the classical longitudinal dynamics balance,

$$P_{wheel} = [m_v a + m_v g C_r \cos \alpha + \frac{1}{2} \rho_{air} C_d A_f v^2 + m_v g \sin \alpha] v, \quad (3)$$

in which the four terms represent the inertial force, the rolling resistance, the aerodynamic drag and the grade force, respectively; for the WLTP cycle a level road,  $\alpha = 0$ , is assumed. The power demanded on the DC bus is then obtained from Eq. (3) through the drivetrain efficiency in a direction-dependent manner,

$$P_{load}(t) = \begin{cases} P_{wheel}(t) / \eta_{dt}, & P_{wheel}(t) \geq 0 \text{ (traction)}, \\ P_{wheel}(t) \eta_{dt}, & P_{wheel}(t) < 0 \text{ (regenerative braking)}, \end{cases} \quad (4)$$

since the drivetrain losses increase the electrical demand during traction and reduce the recoverable power during braking. Eq. (4) is evaluated over the entire cycle to produce the load trajectory used by all strategies.

### 3.3.2. PEMFC stack model: Mann–Amphlett GSSEM

In the GSSEM, the voltage of a single cell is written as the thermodynamic potential reduced by three loss terms [6],

$$V_{FC} = E_{Nernst} - V_{act} - V_{ohm} - V_{con}. \quad (5)$$

The Nernst potential in Eq. (5) is evaluated as

$$E_{Nernst} = 1.229 - 8.5 \times 10^{-4} (T - 298.15) + 4.308 \times 10^{-5} T (\ln p_{H_2}^* + \frac{1}{2} \ln p_{O_2}^*), \quad (6)$$

where  $T$  is the stack temperature in kelvin and  $p_{H_2}^*$ ,  $p_{O_2}^*$  are the partial pressures of the reactants at the catalyst interfaces. The activation overvoltage follows the semi-empirical Tafel-type expression

$$V_{act} = -[\xi_1 + \xi_2 T + \xi_3 T \ln C_{O_2}^* + \xi_4 T \ln i], \quad (7)$$

in which  $\xi_1, \dots, \xi_4$  are the stack-specific semi-empirical coefficients and  $C_{O_2}^*$  is the dissolved oxygen concentration. The ohmic loss is expressed through the membrane and contact resistances as

$$V_{ohm} = i(R_M + R_C), R_M = \rho_M \frac{l}{A}, \quad (8)$$

where the membrane specific resistivity  $\rho_M$  depends on the effective water content  $\lambda$ , the current density and the temperature according to the correlation of Mann et al. [6]. The concentration loss caused by reactant depletion at high current densities is modelled as

$$V_{con} = -\beta \ln \left( 1 - \frac{J}{J_{max}} \right), \quad (9)$$



with  $J = i / A$  the current density and  $\beta$  an empirical coefficient, *following the form adopted in [7]*. The stack voltage and the gross stack power follow from Eq. (5) as

$$V_{stack} = N_{cell} V_{FC}, P_{fc,gross} = V_{stack} I, \quad (10)$$

with  $N_{cell} = 330$  cells and an active area of  $A = 300 \text{ cm}^2$  for the Mirai II FCB130 stack [17]. The model is used under an isothermal assumption ( $T = 343 \text{ K}$ ), *since the stack temperature evolves on a time scale of minutes and is regulated by a dedicated cooling controller, whereas the EMS acts on a time scale of seconds; the influence of thermal management and ageing on EMS performance is investigated in [20] and is left outside the present scope*. The coefficient set is initialised with the values characterised by Mann et al. [6]; within Graduation Project II, the coefficients will be re-identified for the Mirai II stack from the published polarisation data [17] by metaheuristic optimisation, following the methodology surveyed in [7].

For the EMS, the internal electrochemical variables are unnecessarily detailed; what the controller requires is the hydrogen mass flow rate associated with a gross stack power. The efficiency-map form

$$\dot{m}_{H_2} = \frac{P_{fc,gross}}{\eta_{fc}(P_{fc,gross}) L H V_{H_2}} \quad (11)$$

is therefore adopted, where  $L H V_{H_2} = 120 \text{ MJ/kg}$  is the lower heating value of hydrogen and the stack efficiency map  $\eta_{fc}(\cdot)$  is derived offline from Eqs. (5)–(10) and stored as a one-dimensional look-up table. The indicative values for the Mirai II stack are a peak efficiency of approximately 0.70 near 12 kW gross power and approximately 0.54 at maximum power [17]; the strong dependence of efficiency on the operating point is precisely what renders the power split decision non-trivial.

### 3.3.3. Battery model: 1-RC equivalent circuit with hysteresis

The battery pack consists of 960 A123 ANR26650M1B  $\text{LiFePO}_4$  cells in a 96s10p configuration, providing a nominal voltage of 316.8 V, a capacity of  $Q_{pack} = 25 \text{ Ah}$  and a nominal energy of 7.92 kWh. Following the systematic comparison of Nejad et al. [11], the 1-RC + Hysteresis equivalent circuit model is adopted. The relaxation voltage of the single RC branch obeys the discrete state equation

$$V_{RC1}(k+1) = a V_{RC1}(k) + b i_k, a = e^{-\Delta t / \tau_1}, b = R_1(1 - a), \quad (12)$$

with the time constant  $\tau_1 = R_1 C_1 \approx 180 \text{ s}$ . The state of charge evolves by Coulomb counting,

$$SoC(k+1) = SoC(k) - \frac{\eta_i \Delta t}{Q_{pack}} i_k, \quad (13)$$

where  $\eta_i=0.995$  is the coulombic efficiency and the current is positive in discharge. The terminal voltage is assembled from the open-circuit voltage, the ohmic drop and the relaxation state of Eq. (12) as

$$V_k = V_{OC}(SoC_k, direction_k) - i_k R_s - V_{RC1}(k). \quad (14)$$

Because the open-circuit-voltage curve of the LFP chemistry is nearly flat over the operating window, the open-circuit voltage in Eq. (14) is treated as a dynamic state that relaxes towards direction-dependent charge and discharge reference curves according to the hysteresis law of Huria et al. [12],

$$\frac{dV_{OC}}{dSoC} = \frac{dV_{OC}^{ref}}{dSoC} + m_{hyst}(V_{OC}^{ref} - V_{OC}), \quad (15)$$

where the reference curve  $V_{OC}^{ref}$  is the charge branch when the SoC increases and the discharge branch otherwise, each branch being fitted by an eighth-order polynomial in SoC [11], and  $m_{hyst}=150$  governs the transition rate. The cell parameters  $R_s=8\text{ m}\Omega$ ,  $R_1=6\text{ m}\Omega$  and  $C_1=30000\text{ F}$  are taken from [11] and scaled to pack level by the series/parallel arrangement; the RC time constant is invariant under this scaling. The SoC operating window is constrained to  $[0.20, 0.95]$  to limit degradation, the trip starts from the fully charged plug-in condition  $SoC_0=0.90$ , and the charge-depleting target at the end of the trip is  $SoC_{target}=0.25$ .

In accordance with the two-level modelling practice of [14], the full model of Eqs. (12)–(15) serves as the plant model in the Graduation Project II simulator, whereas the EMS optimisation layer internally employs the simplified internal-resistance (Rint) model, retaining only the SoC state.

### 3.4. Power Flow Model

The two conversion chains that connect the decision variable to the physical consumption quantities are assembled as follows. On the fuel cell side, the bus power is referred back to the gross stack power through the converter efficiency and the auxiliary (compressor, cooling, pumping) consumption,

$$P_{fc,net} = \frac{P_{dc}}{\eta_{dc}}, P_{fc,gross} = P_{fc,net} + P_{aux}, \quad (16)$$

with  $\eta_{dc}=0.95$  and  $P_{aux}/P_{fc,gross} \approx 0.10\text{--}0.13$  [17]; the gross power of Eq. (16) is then converted to the hydrogen mass flow rate by Eq. (11). On the battery side, the terminal power  $P_{bat}=V_{bat}I_{bat}$  is combined with the terminal voltage equation, Eq. (14), to yield a quadratic equation in the current, whose physically meaningful root is

$$I_{bat} = \frac{(V_{OC} - V_{RC1}) - \sqrt{(V_{OC} - V_{RC1})^2 - 4R_s P_{bat}}}{2R_s}, \quad (17)$$

and the requirement that the discriminant in Eq. (17) remain non-negative naturally encodes the power capability limit of the pack. Substituting Eq. (17) into the Coulomb counting law gives the complete state dynamics of the optimal control problem,

$$SoC_{k+1} = SoC_k - \frac{\eta_i I_{bat}(P_{bat,k}, SoC_k) \Delta t}{Q_{pack}}, \quad (18)$$

so that the causal chain of the model reads  $P_{dc} \rightarrow P_{bat}$  by Eq. (2),  $P_{bat} \rightarrow I_{bat}$  by Eq. (17) and  $I_{bat} \rightarrow SoC$  by Eq. (18).

A noteworthy property of the assembled efficiency chain is that the grid-to-wheel efficiency of the battery path ( $\approx 0.87$ ) considerably exceeds the tank-to-wheel efficiency of the fuel cell path at its peak ( $\approx 0.61$ ). Energetically, battery usage is therefore always preferable; the optimal decision, however, is taken on a monetary rather than an energetic basis, which motivates the cost function derived next.

### 3.5. Derivation of the Cost Function

#### 3.5.1. Hydrogen consumption cost

With the hydrogen price  $M_{H_2}$  expressed in €/kg, the hydrogen cost accumulated in one sampling interval follows from Eqs. (11) and (16) in explicit dependence on the decision variable,

$$C_{fc,k} = M_{H_2} \cdot \frac{P_{dc,k} / \eta_{dc} + P_{aux,k}}{\eta_{fc}(\cdot) LHV_{H_2}} \cdot \Delta t, \quad (19)$$

which is non-negative for every admissible decision and independent of the battery state.

#### 3.5.2. Electricity consumption cost

In the plug-in charge-depleting scenario, the energy drawn from the battery originates from the grid, not from hydrogen; the classical equivalence-factor interpretation of ECMS is therefore replaced by the direct monetary valuation of grid electricity, following Xu et al. [4] and the plug-in literature [18], [19]. Energy passing through the battery suffers losses twice, once during charging and once during discharging, and the two power flow directions must be weighted differently. This is captured by the direction-dependent round-trip efficiency

$$\eta_{eff}(P_{bat}) = (\eta_{dis} \eta_{chg})^{sgn(P_{bat})}, \quad (20)$$

so that during discharge ( $P_{bat} > 0$ ) the grid energy originally invested per unit of bus energy exceeds unity, while during charging ( $P_{bat} < 0$ ) only the recoverable fraction of the absorbed

energy is credited. With the electricity price  $M_{ele}$  converted to SI units, the per-step electricity cost reads

$$C_{bat,k} = M_{ele} \cdot \frac{P_{bat,k}}{\eta_{eff}(P_{bat,k}, SoC_k)} \cdot \Delta t, \quad (21)$$

which, unlike Eq. (19), depends on both the decision and the state and becomes negative whenever the battery is being charged, thereby crediting recuperated energy at its grid-equivalent value. The efficiencies  $\eta_{dis}$  and  $\eta_{chg}$  are not constants; they are derived in closed form from the ratio of terminal to open-circuit voltage of the battery model and will be tabulated offline as functions of SoC and power for the simulator. The initial plug-in charge itself does not enter the cost functional; it is represented by the boundary conditions  $SoC_0 = 0.90$  and  $SoC_{target} = 0.25$ .

The price parameters are fixed by Germany 2026 market scenarios [23], [24], summarised in Table 3; the default scenario is used for the main results and the remaining scenarios are reserved for the sensitivity analysis of Graduation Project II. It is noted that, in the default scenario, grid electricity is cheaper than hydrogen on a lower-heating-value basis, which creates the theoretical expectation that the optimal strategy exploits the battery maximally within its constraints.

Table 3. Germany 2026 price scenarios for hydrogen (700 bar) [23] and residential electricity [24].

| Scenario | $M_{H_2}$ (€/kg) | $M_{ele}$ (€/kWh) |
|----------|------------------|-------------------|
| High     | 13.85            | 0.40              |
| Default  | 11.00            | 0.35              |
| Low      | 8.00             | 0.32              |

### 3.5.3. Total cost functional

The running cost of one step combines Eqs. (19) and (21),

$$L_k(x_k, u_k) = C_{fc,k}(u_k) + C_{bat,k}(x_k, u_k). \quad (22)$$

Imposing the terminal SoC target as a hard constraint may render the problem infeasible; it is therefore imposed as a soft quadratic penalty,

$$\Phi(SoC_N) = \gamma (SoC_N - SoC_{target})^2, \quad (23)$$

with the tuning weight initialised at  $\gamma = 1000$  €, a value chosen by dimensional reasoning so that a unit SoC deviation is penalised at a magnitude far exceeding the expected trip cost of 5–15 €. The total cost functional to be minimised is obtained by summing the running cost of Eq. (22) over the horizon and adding the terminal penalty,

$$J(\{u_k\}_{k=0}^{N-1}) = \sum_{k=0}^{N-1} L_k(x_k, u_k) + \Phi(SoC_N), \quad (24)$$

and Eq. (24) constitutes the principal mathematical result of Graduation Project I. Component degradation costs of the fuel cell and the battery are deliberately excluded from Eq. (24) and deferred to future work, in line with [1]; the load-ramp limitation introduced by the BLFS layer in Section 3.6.5 nevertheless protects the fuel cell durability indirectly.

### 3.6. Optimal Control Problem and EMS Strategy Formulations

#### 3.6.1. Complete problem statement

Collecting the cost functional, the state dynamics and the physical limits, the complete optimal control problem is stated as

$$\min_{\{P_{dc,k}\}_{k=0}^{N-1}} J = \sum_{k=0}^{N-1} L_k(SoC_k, P_{dc,k}) + \gamma(SoC_N - SoC_{target})^2, \quad (25)$$

subject to the state dynamics of Eq. (18), the power balance of Eq. (2), the initial condition  $SoC_0 = 0.90$ , and the operating limits

$$0 \leq P_{dc,k} \leq P_{dc}^{\max}, P_{bat,k}^{\min} \leq P_{bat,k} \leq P_{bat}^{\max}, SoC_{\min} \leq SoC_k \leq SoC_{\max}, \quad (26)$$

in which the battery power bounds are SoC-dependent as encoded by the discriminant of Eq. (17). The problem in Eqs. (25)–(26) involves  $N = 3600$  scalar decisions and a one-dimensional state, a size that is tractable offline by DP and in real time by the remaining strategies. The four solution strategies formulated below will be implemented and compared in Graduation Project II.

#### 3.6.2. Dynamic Programming

DP computes the global optimum of Eqs. (25)–(26) by the Bellman backward recursion over the cost-to-go function,

$$V_k(SoC_k) = \min_{P_{dc,k}} \{L_k(SoC_k, P_{dc,k}) + V_{k+1}(SoC_{k+1})\}, V_N = \Phi(SoC_N), \quad (27)$$

evaluated on a discretised state–decision grid ( $\Delta SoC = 0.005$ ,  $\Delta P_{dc} = 1$  kW), following the implementation structure of Sundström and Guzzella [15] and the theoretical framework of Bertsekas [16]. Since the recursion in Eq. (27) presupposes knowledge of the entire load trajectory, DP is not implementable in a real vehicle and serves exclusively as the optimality benchmark against which the real-time strategies are measured.

### 3.6.3. Adaptive ECMS

The real-time counterpart of the optimal solution is constructed from Pontryagin's Minimum Principle, according to which the optimal decision minimises, at every instant, the Hamiltonian

$$H(SoC, P_{dc}, \lambda, t) = L(SoC, P_{dc}) + \lambda \dot{SoC}, \quad (28)$$

where  $\lambda$  is the co-state [2], [4]. In the adaptive ECMS implementation, the role of the co-state is taken by an equivalence factor  $s(t)$  which is updated online by a proportional-integral law on the SoC tracking error, following Gao et al. [5],

$$s(t) = s_0 + m e_{SoC}(t) + n \int_0^t e_{SoC}(\tau) d\tau, e_{SoC} = SoC_{ref}(t) - SoC(t), \quad (29)$$

where the reference trajectory  $SoC_{ref}(t)$  decreases linearly from 0.90 to 0.25 over the trip, in accordance with the charge-depleting character of the plug-in operation [4]. The adaptation in Eq. (29) is essential in the plug-in case: Xu et al. proved that a constant equivalence factor can drive the solution to a degenerate boundary strategy [4], whereas the feedback law continually steers the SoC towards its reference. At each sampling instant the Hamiltonian of Eq. (28) is evaluated on a discrete decision grid and the minimising  $P_{dc}$  is applied.

### 3.6.4. Model Predictive Control

MPC solves, at every instant  $k$ , the finite-horizon restriction of Eqs. (25)–(26),

$$\min_{\{P_{dc,j}\}_{j=k}^{k+N_p-1}} \sum_{j=k}^{k+N_p-1} L_j(SoC_j, P_{dc,j}) + \gamma_{MPC} (SoC_{k+N_p} - SoC_{ref}(t_{k+N_p}))^2, \quad (30)$$

subject to the dynamics of Eq. (18) and the limits of Eq. (26), applies the first element of the optimal sequence and repeats the optimisation at the next instant (receding horizon). The prediction horizon is initialised at  $N_p = 15$  s. The future load over the horizon is initially predicted by the simplest constant-load assumption; the problem in Eq. (30) will be transcribed to a quadratic programme and solved with CasADi/IPOPT in Graduation Project II, where the discontinuity of the sign function in Eq. (20) will be smoothed by a hyperbolic-tangent approximation.

### 3.6.5. Boundary Layer Surface Following

BLFS is not an independent EMS but a lightweight lower-level protection layer operating beneath A-ECMS or MPC. It defines a hysteresis band of half-width 0.05 around the SoC reference trajectory: above the band the fuel cell is held at minimum power so that the battery is depleted, below the band the fuel cell is held at its peak-efficiency operating point

so that the battery is recharged, and inside the band the command of the upper-level strategy is passed through. In addition, the fuel cell load ramp is limited to

$$\left| \frac{d P_{fc}}{d t} \right| \leq 5 \text{ kW} / \text{s}, \quad (31)$$

which prevents gas starvation and membrane fatigue caused by rapid transients; the constraint in Eq. (31) thereby protects fuel cell durability without entering the cost functional explicitly. The qualitative comparison of the four strategies is summarised in Table 4.

Table 4. Qualitative comparison of the four EMS strategies.

| Property             | DP             | A-ECMS                   | MPC                  | BLFS              |
|----------------------|----------------|--------------------------|----------------------|-------------------|
| Information required | Full cycle     | Instantaneous + SoC ref. | Finite horizon       | Instantaneous SoC |
| Optimality           | Global         | Near-optimal (PMP)       | Local                | None (rule-based) |
| Real-time capable    | No             | Yes                      | Yes                  | Yes               |
| Computational load   | High (offline) | Low                      | Medium               | Very low          |
| Role                 | Benchmark      | Production candidate     | Production candidate | Protection layer  |

### 3.7. Current Status of the Study and Outline of Graduation Project II

*Within the present term, the literature survey has been completed, all architectural selections have been fixed on the basis of primary sources, the component models have been formulated, the plug-in operational cost function has been derived, and the complete optimal control problem, Eqs. (25)–(26), has been stated together with the formulation of the four solution strategies, Eqs. (27)–(31).*

*Graduation Project II will be devoted to the numerical solution of this formulation. First, a two-layer simulation environment will be implemented in Python: a high-fidelity plant layer, combining the Mann-Amphlett stack model with the full 1-RC + Hysteresis battery model, and a simplified controller layer based on the efficiency look-up table and the Rint model. Second, the semi-empirical stack coefficients will be identified for the Mirai II from the published polarisation data [17] by metaheuristic optimisation (PSO/GA), and the efficiency map of Eq. (11) will be generated. Third, the four strategies will be implemented in sequence: the DP benchmark by backward recursion, A-ECMS with the PI-adapted equivalence factor, MPC transcribed to a quadratic programme and solved with CasADi/IPOPT, and the BLFS protection layer integrated beneath the real-time strategies.*

**Finally, the strategies will be compared over the reference trip in terms of total operating cost (€/100 km), hydrogen and electricity consumption, terminal SoC accuracy, fuel cell quasi-stability and per-step computation time, and a sensitivity analysis will be performed over the price scenarios of Table 3 and the penalty weight of Eq. (23).**

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